Hydrogen bromide is more difficult to oxidise than HI. HBr reduces slowly H_2SO_4 into SO_2

$$2HBr + H_2SO_4 \longrightarrow 2H_2O + Br_2 + SO_2$$

But hydrogen iodide and ionic iodides are rapidly reduced by H_2SO_4 into H_2S and not into SO_2.

$$8HI + H_2SO_4 \longrightarrow 4H_2O + 4I_2 + H_2S$$

Reducing property of hydrogen iodide can be also explained by using its reaction with alcohols into ethane. It converts nitric acid into nitrous acid and dinitrogen dioxide into ammonium.

Hydrogen chloride is unaffected by concentrated sulphuric acid but affected by only strong oxidising agents like MnO_2, potassium permanganate or potassium chloride.

To summarize the trend,

Table 3.5

Property	Order
Reactivity of hydrogen	Decreases from fluorine to iodine
Stability	Decreases from HF to HI
Volatility of the hydrides	HF < HI < HBr < HCl
Thermal stability	HF > HCl > HBr > HI
Boiling point	HCl < HBr < HI < HF
Acid strength	Increases from HF to HI

3.3.4 Inter halogen compounds:

Each halogen combines with other halogens to form a series of compounds called inter halogen compounds. In the given table of inter halogen compounds a given compound A is less electronegative than B.

Table 3.6

AB	AB_3	AB_5	AB_7
ClF	ClF_3	IF_5	IF_7
BrF	BrF_3	BrF_5	
IF	IF_3		
BrCl	ICl_3		
ICl			
IBr			

Properties of inter halogen compounds:

i. The central atom will be the larger one

ii. It can be formed only between two halogen and not more than two halogens.

XII U3-P-block.indd 89 16-12-2022 18:37:17